Project Proposal: Robustness in Generative Spoken Language Modeling

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1 Summary of the Reference Article

The project is based on the paper *"Augmentation Invariant Discrete Representation for Generative Spoken Language Modeling"* (Gat et al., ACL 2023).

Context & Problem: Generative Spoken Language Modeling (GSLM) relies on discrete units derived from self-supervised models (e.g., HuBERT, wav2vec 2.0) via k-means quantization. The authors identify that these representations lack robustness to signal variations that preserve linguistic content, such as time-stretching, pitch-shifting, reverberation, and additive noise.

Proposed Solution: The paper introduces a metric, *Unit Edit Distance* (UED), to quantify the stability of discrete sequences under augmentation. To improve robustness, they propose a pseudo-labeling method. This involves training a lightweight quantizer (MLP) to map augmented audio representations to the discrete units of the original "clean" audio using Connectionist Temporal Classification (CTC) loss. This aligns the augmented signal with the clean discrete sequence, effectively distilling robustness into the new quantizer.

Results: The method significantly reduces UED and improves performance on downstream tasks, including zero-shot metrics (ABX, sWUGGY, sBLIMP) and speech-to-speech translation, outperforming standard k-means baselines.

2 Project Objectives

Our goal is to reproduce the findings of Gat et al. to establish a strong baseline, and subsequently extend the evaluation to modern neural codecs and cross-lingual settings.

2.1 Phase 1: Reproduction & Benchmarking

We aim to validate the claims regarding the fragility of current SSL units.

- **Baseline Evaluation:** We will use pre-trained HuBERT and Wav2Vec 2.0 models to extract discrete units. We will implement the *Unit Edit Distance* (UED) and the ABX metric to measure robustness against the four primary augmentations: time-stretch, pitch-shift, reverberation, and noise.
- **Robust Quantization:** We will implement the proposed pseudo-labeling training pipeline. This involves freezing the upstream encoder and training the robust MLP quantizer using the CTC loss between clean and augmented audio views.
- **Verification:** We verify if the re-trained quantizer achieves the reported improvements in UED and ABX scores.

2.2 Phase 2: Extensions

Building upon the reproduction, we propose the following extensions to test the generalizability of the method:

- **Cross-lingual Robustness:** The original paper trains the robust quantizer primarily on English (LibriSpeech). We propose to evaluate if a quantizer trained to be robust on English audio generalizes to other languages (e.g., French or Spanish). Does "robustness to reverb" learned on English transfer to preserving phonemes in French?

- **Additional Augmentations:** We plan to expand the augmentation set beyond the original four. Potential additions include codec compression artifacts (simulating packet loss or low bitrate) to see if the model can learn invariance to compression noise.
- **Modern Encoders (EnCodec):** While the paper focuses on SSL models (HuBERT), neural codecs like EnCodec are becoming the standard for generation (e.g., VALL-E, MusicGen). EnCodec uses Residual Vector Quantization (RVQ) rather than k-means. We investigate whether EnCodec codes suffer from similar robustness issues and if the paper's CTC-alignment method can be adapted to fine-tune or robustify EnCodec's discrete codes.